# Answer 04

Here first, we are calculating TPR and FPR using the confusion matrices.

| Hypothesis | TPR | FPR |
|---|---|---|
| H1 | 29/31 = 0.9355 | 1/14 = 0.0714 |
| H2 | 29/30 = 0.9667 | 3/15 = 0.2000 |
| H3 | 27/30 = 0.9000 | 3/15 = 0.2000 |

## Case 01 - Equal cost

$$\text{Distance} = \sqrt{(FPR^2 + (1 - TPR)^2)}$$

| Hypothesis | Distance |
|---|---|
| H1 | 0.096 |
| H2 | 0.203 |
| H3 | 0.224 |

H1 is the best classifier because it has the smallest distance from the perfect classifier.

## Case 02 - False Positive cost 4 times more

$$\text{Weighted Distance} = \sqrt{((4 \times FPR)^2 + (1 - TPR)^2)}$$

| Hypothesis | Distance |
|---|---|
| H1 | 0.292 |
| H2 | 0.801 |
| H3 | 0.806 |

H1 is again the best classifier as it has the smallest weighted distance.

Hence, H1 performs best in both the cases.